UIL High School Science Contest
ANSWER KEY
Invitational A • 2016

1.	C	21.	E	41.	D
2.	B	22.	C	42.	B
3.	D	23.	A	43.	E
4.	C	24.	D	44.	A
5.	B	25.	A	45.	A
6.	B	26.	B	46.	C
7.	D	27.	B	47.	C
8.	A	28.	D	48.	A
9.	A	29.	C	49.	E
10.	C	30.	E	50.	D
11.	C	31.	A	51.	B
12.	D	32.	C	52.	C
13.	B	33.	E	53.	D
14.	C	34.	B	54.	E
15.	C	35.	D	55.	A
16.	C	36.	B	56.	E
17.	B	37.	C	57.	B
18.	A	38.	E	58.	C
19.	C	39.	A	59.	D
20.	D	40.	D	60.	D

PHYSICS KEY for Science Contest • Invitational A Meet • 2016

41. (D) page 4: "…what statement would contain the most information in the fewest words? I believe it is the atomic hypothesis (or the atomic fact, or whatever you wish to call it) that all things are made of atoms…"

42. (B) page 12: "Therefore, since those that leave have more energy than the average, the ones that are left have less average motion than before. So the liquid gradually cools if it evaporates."

43. (E) page 29: "…if everything attracted everything else instead of likes repelling, so that there were no cancellations, how much force would there be? There would be a force of three million tons between the two!"

44. (A) page 34: "…If they were in the nucleus, we would know their position precisely, and the uncertainty principle would then require that they have a very large (but uncertain) momentum, i.e., a very large kinetic energy."

45. (A) When the hydrogen fuel is exhausted, the star's core collapses until the core increases to a temperature of about 10^8 K. Then the triple-alpha process can begin fusing Helium into Carbon.

46. (C) $x = x_0 + v_0 t + {}^1\!/_2\, at^2$

    $900 = 0 + (200)(3.5) + {}^1\!/_2\, a(3.5)^2$ gives $a = 32.7\ {}^m\!/_{s^2}$

    Then use $v = v_0 + at$

    $v = 200 + (32.7)(3.5) = 314\ {}^m\!/_s$

47. (C) Conservation of momentum: $m_1 v_1 = m_2 v_2 + m_1 v_f$

    so, $(m_1 v_1 - m_2 v_2)/_{m_1} = v_f$

48. (A) Parallel Branches: $V_0 = V_1 = V_2$ so the voltage across the upper branch V_2 = 12.0 V

    Resistance of the upper branch: resistors in series $R = R_1 + R_2$, so R = 60 + 30 = 90 Ohms

    Current in the upper branch: V_2/R = 12/90 = 0.13333 A

    Voltage across the 60 Ohm: V = IR = (0.133)*(60) = 8.0 V

49. (E) Forces in the y-direction: N-mg = 0, so N = mg = (40)*(9.8) = 392 N

    Forces in the x-direction: F - f = ma, and f = μN = (0.22)(392) = 86.2 N

    So a = (160 – 86.2)/40 = 1.8 m/s^2

50. (D) Period: $T = 2\pi\sqrt{m/k}$ = 1.35, so $k = (m)(2\pi/T)^2$ = $(0.1\ kg)(2\pi/1.35)^2$ = 2.17 N/m

    Energy: $E = \frac{1}{2}kA^2 = \frac{1}{2}(2.17)(0.15\ m)^2 = 0.0244\ J = \frac{1}{2}mv_{max}^2$

    So maximum velocity: $v_{max} = \sqrt{(2)(0.0244)/(0.1\ kg)}$ = 0.698 m/s = 69.8 cm/s

51. (B) efficiency: $= W/Q_{in} = 0.37$, so : $Q_{in} = (W/e) = 162\ J$

    $Q_{out} = Q_{in} - W = 162 - 60 = 102\ J$

52. (C) At resonance, the circuit is resistive (not capacitive, nor inductive); power output is maximum; and the current and voltage are in phase, so the phase angle is zero.

53. (D) Vector electric field cannot be zero between two opposite charges; it will be zero at a point beyond the smaller charge. Let the distance from the negative charge be X, then the distance from the positive charge is X – r.

    Then: $kQ/(X - r)^2 - 2kQ/(X)^2 = 0$, leading to $1/(X - r)^2 = 2/X^2$

    This produces the quadratic equation: $X^2 - 4Xr + 2r^2 = 0$, the solution of which is $X = 2r + \sqrt{2}r$

54. (E) For a charged particle in a magnetic field: $r = mv/qB$, so $m = qBr/v$

    m = (1.6 x 10^{-19} C)(0.12 T)(0.327 m)/(2.5 x 10^5 m/s) = 2.5 x 10^{-26} = 15 amu

55. (A) f = R/2 = 18.0 cm. $1/p + 1/q = 1/f$, so $1/13 + 1/q = 1/18$, which gives an image location of q = -46.8 cm. Magnification M = -q/p = -(-46.8)/13 = 3.6. The image height is h' = (3.6)(15) = 54 cm

ANSWER KEY

56. (E) Use Energy: initially in the form of elastic potential energy: $E = \frac{1}{2}kx^2 = (0.5)(50)(.15)^2 = .5625\,J$

    This energy is dissipated by the work done on the block by friction. $E = fd = .5625 = f(1.2)$

    So the frictional force is f = 0.469 N. From the force diagram, the normal force is given by

    N = mg = (0.55)(9.8) = 5.4 N. Then the coefficient of friction μ = f/N = (0.469)/(5.4) = 0.087

57. (B) Californium-251 has atomic number Z = 98 and atomic mass M = 251. Four alpha (Z = 2, M = 4), two Beta-minus (Z = -1, M = 0), and two gamma (Z = 0, M = 0) emissions lowers the atomic number and atomic mass as follows: Z = 98 – 4(2) + 2(1) = 92 → Uranium. M = 251 – 4(4) = 235

58. (C) The neutral pion consists of a quark and its corresponding anti-quark (uū, for example). The quark anti-quark pair annihilate and produce gamma rays (photons, the electromagnetic exchange bosons). This type of matter-antimatter decay is mediated entirely by the electromagnetic force.

59. (D) The intensity of the uniformly expanding radio waves is given by:

    $I = P/(4\pi r^2) = (1000)/(4\pi * 400^2) = 4.97 \times 10^{-4}$ W/m^2. Intensity is related to electric field by:
    $I = \frac{1}{2}c\varepsilon_0 E^2$, so $E = \sqrt{2I/c\varepsilon_0} = 0.61$ N/C

60. (D) Wavelengths in nanometers are given by $\lambda = 1240/\Delta E$ where ΔE (energy level difference) is given in eV. ΔE values for this atom are: 6.14 eV, 4.37 eV, 3.42 eV, 2.72 eV, 1.77 eV, and 1.65 eV. This gives wavelengths of 202 nm, 284 nm, 363 nm, 456 nm, 701 nm, and 752 nm.